

Transformer-based classification of variable stars in TESS light curves

Multi-resolution views and fine-tuned time series foundation models

A. Mateides¹ and P. Škoda^{1,2}

¹ Faculty of Information Technology, Czech Technical University in Prague, Thákurova 9, 160 00 Prague 6, Czech Republic

² Astronomical Institute of the Czech Academy of Sciences, Fričova 298, 251 65 Ondřejov, Czech Republic

Received —; accepted —

ABSTRACT

Context. Surveys such as OGLE, Kepler, ZTF, and TESS deliver light curves in numbers that make manual classification of variable stars impractical, and Gaia DR4 and PLATO will soon add more.

Aims. We assess whether transformers are a viable backbone for variable-star classification from raw TESS photometry, and whether pretrained time series foundation models (TSFMs) and large language models (LLMs) can be fine-tuned to outperform both from-scratch transformers and a convolutional baseline.

Methods. We used 3347 manually classified 2-min cadence light curves from the northern TESS continuous viewing zone, split into Task A (variable versus non-variable) and Task B (pulsating, rotating, eclipsing). Each light curve was cut into fixed-length segments, and each segment represented by four time-domain views at 2, 8, 32, and 128 min effective cadence, a phase-folded view, and scalar features. A multi-branch encoder processes each view in its own transformer branch using statistical patch embeddings, then fuses the branches linearly. Interchangeable backbones host transformers trained from scratch, a Chronos-2 TSFM, or a Qwen 2.5 0.5B LLM, the latter two fine-tuned with LoRA. Predictions are aggregated to the object (TIC) level by mean class probability.

Results. Across 16 experiments, fine-tuned Chronos-2 was the only configuration beating the convolutional baseline on both tasks, reaching validation TIC-level macro F_1 of 0.926 (Task A) and 0.950 (Task B) against 0.906 and 0.923 for the baseline, and 0.893 and 0.880 on the test set. The Task B loss is driven almost entirely by five atypical eclipsing systems. Multi-channel input and segmentation, not the backbone, account for most of the gain over single-view models.

Conclusions. Fine-tuned TSFMs transfer effectively to stellar variability classification despite never having seen astronomical data, and the multi-resolution view representation is the decisive design choice. The pipeline processes $\sim 10^6$ light curves per day on consumer hardware, making it applicable to survey-scale archives and to reducing exoplanet false positives.

Key words. Methods: data analysis – Techniques: photometric – Stars: variables: general – Binaries: eclipsing – Surveys

1. Introduction

Variable stars underpin a large part of observational astrophysics. Their study produced the distance ladder that led to the discovery of the expansion of the Universe (Hubble 1929), and the same photometric time series that reveal intrinsic variability are the raw material of transit-based exoplanet searches.

Progress in photometric instrumentation has produced surveys of unprecedented volume: OGLE (Udalski et al. 2008), Kepler (Borucki et al. 2010), the Zwicky Transient Facility (Bellm et al. 2019), and the Transiting Exoplanet Survey Satellite (TESS, Ricker et al. 2014). Two further large releases are imminent – Gaia DR4 (Gaia Collaboration et al. 2016) and the PLATO mission (Rauer et al. 2025) – and both will deliver light curves in quantities that rule out manual inspection.

Machine learning has become the standard response. Convolutional networks in particular have repeatedly outperformed feature-engineered classifiers built on random forests or clustering. More recently, transformer architectures (Vaswani et al. 2017) have become the dominant family for sequence modelling, and a parallel development – large models pretrained on generic corpora and then adapted to a downstream task – has proven ef-

fective in domains with little labelled data. Both developments are directly relevant here: labelled variable-star catalogues are small relative to the size of modern networks, while unlabelled light curves are abundant.

In this work we ask two questions. First, can a transformer trained from scratch on a few thousand labelled TESS light curves compete with a convolutional baseline? Second, can large pretrained models – a time series foundation model (TSFM) and a large language model (LLM), neither of which has ever seen astronomical data – be fine-tuned into competitive classifiers of stellar variability? We answer both by evaluating 16 configurations of a single modular architecture under an identical training and evaluation protocol.

The paper is organised as follows. Section 2 describes the TESS dataset and the two classification tasks. Section 3 details the preprocessing pipeline, in particular the multi-resolution view representation that makes long light curves tractable for attention-based models. Section 4 presents the architecture, the training objective, and the object-level evaluation protocol. Section 5 reports the experiments, and Section 6 discusses failure modes, throughput, and applicability. Section 7 concludes.

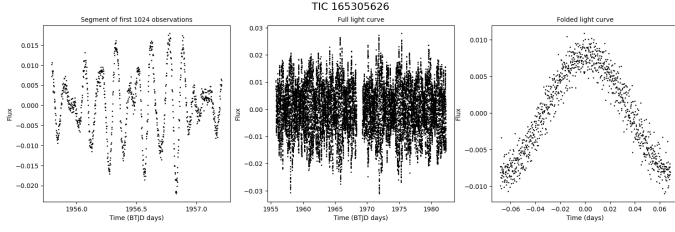


Fig. 1. Raw 2-min cadence light curve of a δ Scuti star. Note the observing gap near the centre of the sector.

2. Data

2.1. TESS light curves

TESS (Ricker et al. 2014) was launched by NASA in 2018. Although its primary mission is the detection of transiting exoplanets, it monitors every target that shows brightness variations, making it a rich source of variable-star photometry. Its four 16-megapixel wide-field cameras produce full-frame images from which individual targets are extracted as target pixel files; fluxes are then integrated to form time series, each identified by a unique TESS Input Catalogue number (TIC) (MAST 2024).

We use light curves recorded at 2-min cadence. Their lengths vary considerably, from roughly 11 000 to 216 000 points. In raw form each record contains only a timestamp t in Barycentric TESS Julian Date, a flux measurement Φ , and its uncertainty Φ_{err} . Figure 1 shows a typical example, including the observing gap that occurs at each spacecraft perigee passage.

2.2. Labelled sample

Our sample is a cleaned subset of the manually labelled catalogue of periodic variable A–F stars in the northern TESS continuous viewing zone published by Skarka et al. (2022), to which we refer for the full classification procedure.

The catalogue contains 3402 light curves annotated with 49 distinct class labels, of which only 13 have more than ten members. We merged these into five superclasses. Objects carrying ambiguous multi-class annotations spanning different superclasses (e.g. ROTM/ELL) and extremely rare classes were discarded; this affected 55 objects and left 3 347 light curves. Table 1 summarises the mapping.

2.3. Hierarchical task definition

Separating variables from non-variables and assigning a variability type to a confirmed variable are related but distinct problems, with very different class balance and very different decision boundaries. We therefore adopt a hierarchical formulation:

Task A Binary classification: *variable* (U, P, R, E merged) versus *non-variable* (N). This task uses all 3 347 objects and is close to balanced (1 729 vs 1 618).

Task B Three-class classification of confirmed variables of known type: *pulsating* (535), *rotating* (262), *eclipsing* (101), for a total of 898 objects. The unspecified-variable class is excluded, since it carries no type information.

Task B is the harder problem: it has roughly a quarter of the objects of Task A and a 5:1 imbalance between its most and least populated classes.

Table 1. Label mapping from the original annotations of Skarka et al. (2022) to the five superclasses used in this work.

Superclass	Original labels	Count	Total
Non-variable (N)	NONE	1 729	1 729
Unspecified variable (U)	VAR	720	720
Pulsating (P)	GDOR	206	535
	DSCT+GDOR	176	
	GDOR+DSCT	86	
	DSCT	67	
Rotating (R)	ROTS	175	262
	ROT	74	
	ROTM	13	
Eclipsing (E)	EA	47	101
	ELL	24	
	EW	17	
	EB	13	
Total			3 347

3. Preprocessing

Raw light curves cannot be fed directly to an attention-based model: they are far too long, they contain gaps, and their flux scales differ by orders of magnitude between targets. We describe the pipeline used to produce model inputs, including the steps that were evaluated and rejected.

3.1. Time standardisation and flux normalisation

Each light curve is shifted to a common origin, $t' = t - t_0$, so that all series start at $t' = 0$. Fluxes are mapped to $[-1, 1]$ by a modified min–max normalisation,

$$\Phi' = 2 \frac{\Phi - \Phi_{\min}}{\Phi_{\max} - \Phi_{\min}} - 1, \quad (1)$$

which centres the series on zero and thus reflects the physical behaviour of a flux oscillating about a mean level. Normalisation to $[0, 1]$ gave statistically indistinguishable results. Because variability class is encoded mainly in light-curve morphology rather than amplitude, discarding the amplitude scale improves generalisation; the amplitude itself is retained separately as a scalar feature (Sect. 3.7).

3.2. Detrending

Detrending is routinely applied to remove instrumental drifts and low-frequency systematics,

$$\Phi' = \Phi - \Phi_{\text{trend}}, \quad (2)$$

with Φ_{trend} estimated by a running median, a Gaussian convolution, or a low-order polynomial fit. We evaluated all three with a range of window sizes and kernel widths. None improved classification performance over the untreated series. The reason is physical: rotational variables are defined precisely by the long-timescale modulation that these filters suppress. Detrending remains valuable for other tasks – eclipse searches in particular – but was removed from our final pipeline.

3.3. Observing gaps

Invalid flux values (NaN, $\pm\infty$) are identified and encoded in a binary validity mask. The mask is applied inside the attention

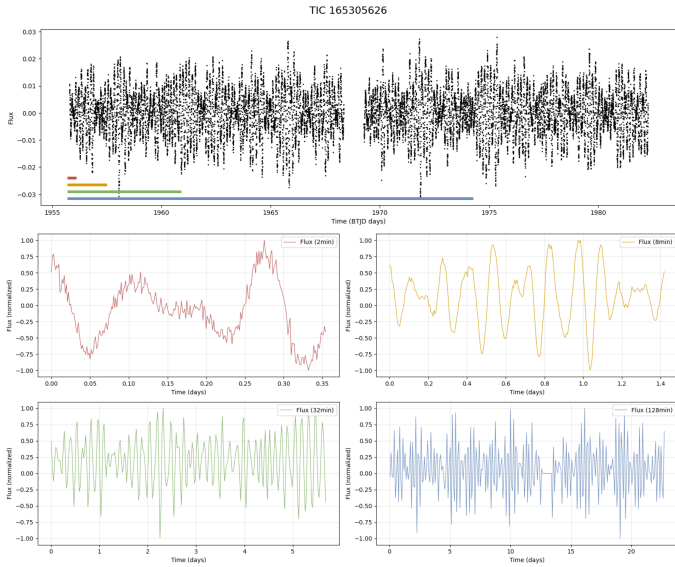


Fig. 2. Four time-domain views of the light curve of a δ Scuti star. Each view contains 256 points sampled at effective cadences of 2, 8, 32, and 128 min. The coloured bars indicate the extent of each view within the original series.

operation, so that invalid positions cannot contribute to any other token. When a CLS token is used for classification no further handling is needed; with mean pooling the mask must be applied a second time so that invalid tokens are excluded from the pooled representation.

3.4. Multiple time-domain views

The cost of self-attention grows quadratically with sequence length (Vaswani et al. 2017), which makes ingesting a full 2-min cadence sector – 10^5 points or more – infeasible. Downsampling to a manageable length, on the other hand, destroys the short-period signal that distinguishes δ Scuti pulsators.

Our solution is to present the model with several views of the same interval at different temporal resolutions. Each view has a fixed length of 256 points, and is obtained by selecting a window at the start of the segment and mean-binning it to that length. We use four resolutions:

- 2 min (native cadence, bin size 1),
- 8 min (bin size 4),
- 32 min (bin size 16),
- 128 min (bin size 64).

A 128-min view of length 256 therefore covers $(128/2) \times 256 = 16\,384$ raw points, i.e. roughly 23 d, while the 2-min view resolves individual pulsation cycles over the first ~ 8.5 h. Since all views start at the same point, the higher-resolution views are strict sub-intervals of the lower-resolution ones (Fig. 2). Together they span timescales from minutes to weeks at constant input length.

3.5. Segmentation

The labelled sample is small – 898 objects with a specified variability type, only 101 of them eclipsing – which severely limits the capacity of network that can be trained. We therefore split each light curve into consecutive segments, each the length of the largest view window (4096 raw points minimum, stride

Table 2. Composition of the two task datasets after preprocessing.

Task	Class	Objects	Segments
A	Non-variable (N)	1 729	57 262
	Variable (V)	1 618	56 093
	Total	3 347	113 355
B	Pulsating (P)	535	17 868
	Rotating (R)	262	9 806
	Eclipsing (E)	101	3 176
	Total	898	30 850

4096). Incomplete trailing segments are retained and masked. This yields 113 355 training segments for Task A and 30 850 for Task B (Table 2), more than an order of magnitude more samples than there are objects.

Segmentation introduces three complications, all of which must be handled explicitly:

- *Leakage.* Train/validation/test splits are made at the TIC level, so all segments of a given star fall in the same subset.
- *Segment imbalance.* Objects yield different numbers of segments, so naive class weights are biased towards classes whose members happen to have longer light curves. This is corrected in the loss (Sect. 4.3).
- *Evaluation.* Segment-level metrics do not answer the scientific question. Predictions are aggregated to the object level (Sect. 4.4).

3.6. Phase-folded view

A Lomb–Scargle periodogram (Lomb 1976; Scargle 1982) is computed for each light curve, the dominant period is extracted, and the series is phase-folded on it and binned to the standard length of 256. This provides a fifth input channel in which periodic signals are coherently stacked, complementing the four time-domain views.

3.7. Scalar features

Three groups of scalar quantities are extracted and supplied through a separate branch: the pre-normalisation amplitude, which restores the physical scale removed by normalisation; the ten strongest periods identified in the periodogram; and their corresponding powers. These carry information that is difficult for a patch-based encoder to recover from a truncated view.

4. Method

4.1. Architecture

The classifier consists of six parallel branches feeding a linear fusion layer (Fig. 3): five encoder-only transformer branches, one per input channel (four time-domain views plus the phase-folded view), and one multilayer-perceptron branch for the scalar features. Each branch can be enabled or disabled independently through the configuration file, which allows any subset of channels to be evaluated without code changes. The transformer blocks themselves are interchangeable, so the same architecture can host encoders trained from scratch or pretrained backbones.

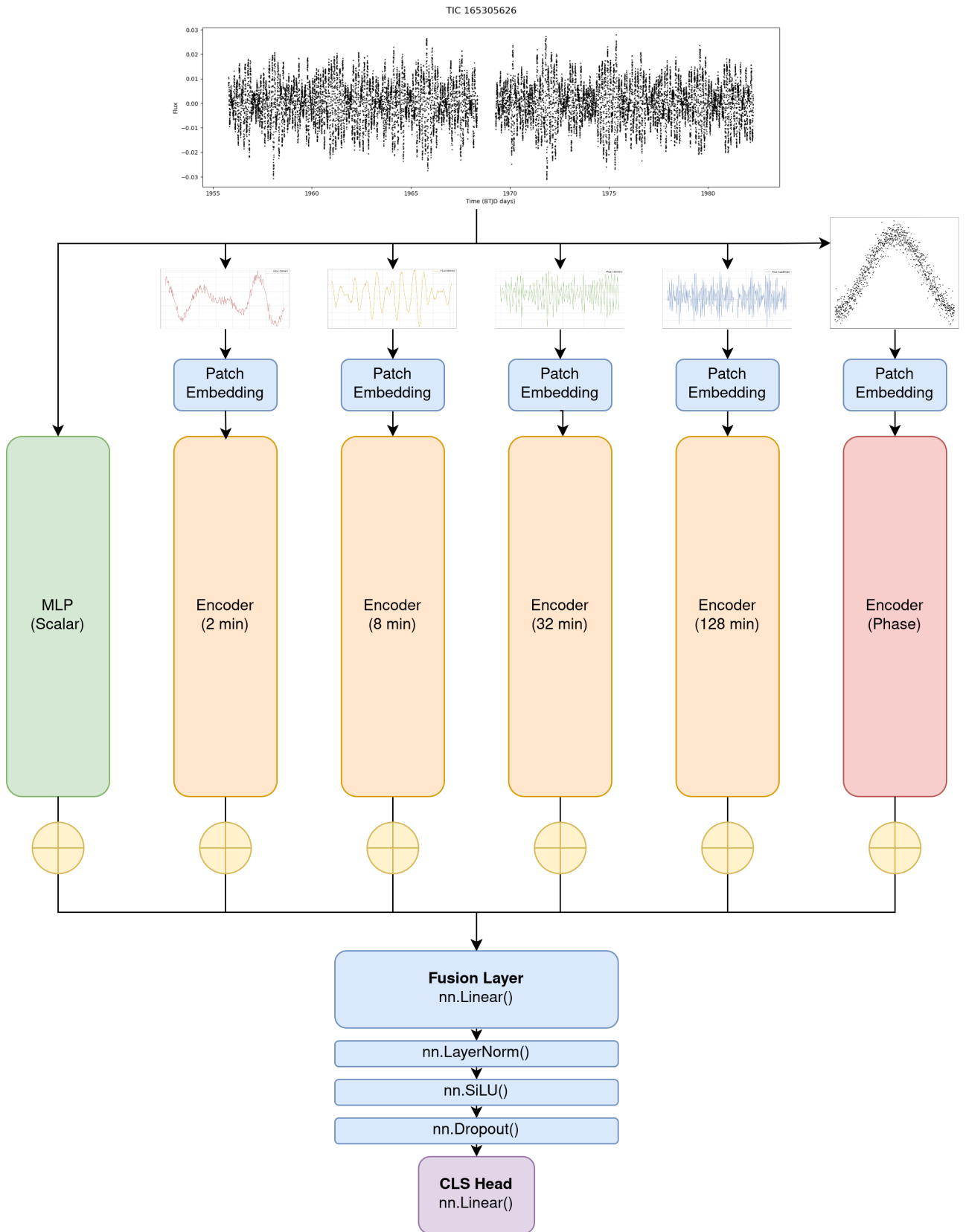


Fig. 3. Multi-branch architecture. Each of the five sequence channels is processed by a separate encoder-only transformer branch; scalar features pass through a small MLP. The branch representations are combined by a linear fusion layer before classification. Any branch can be disabled, allowing arbitrary channel combinations to be tested under an otherwise identical configuration.

4.2. Statistical patch embeddings

Per-point learnable embeddings were tested first, but overfitted readily and made attention costly, since every flux measure-

ment becomes a token. We therefore use patch embeddings, augmented with summary statistics computed from each patch.

An input tensor of shape (L, D) , where L is the view length and D the number of input channels (Φ and Φ_{err}), is reshaped to (N, P, D) for patch size P and $N = L/P$ patches, padding as required. For each patch j and channel d we compute, using only valid positions according to the mask, the mean $\mu_{j,d}$, the standard deviation $\sigma_{j,d}$, the minimum, the maximum, and the slope, approximated as $\text{cov}(\Phi, t)/\text{var}(t)$; a single additional scalar records the fraction of valid points in the patch. The resulting vector of length $5D + 1$ is projected to d_{model} by a two-layer MLP. Relative to raw-flux patch embeddings, this representation improved macro F_1 by 0.05–0.10, the single largest architectural gain we measured.

4.3. Segment- and class-weighted loss

Training minimises a segment-level cross-entropy in which class weights correct for class imbalance and a per-segment weight prevents objects with many segments from dominating. For segment i belonging to object t with n_t segments, the segment weight is $s_i = 1/n_t$, and the batch loss is

$$\mathcal{L} = \frac{\sum_{i=1}^B s_i L_{\text{WCE}}^{(i)}}{\sum_{i=1}^B s_i}, \quad (3)$$

where B is the batch size and $L_{\text{WCE}}^{(i)}$ is the class-weighted cross-entropy of segment i . The objective remains segment-based, but every object contributes approximately equally.

4.4. Object-level aggregation and metrics

The scientific target is the classification of a star, not of a 23-day window of its light curve. For each segment i of object t the model produces logits $\mathbf{z}_{t,i}$ over C classes, which are converted to probabilities by a softmax and averaged over the n_t segments,

$$\bar{p}_{t,c} = \frac{1}{n_t} \sum_{i=1}^{n_t} \text{softmax}(\mathbf{z}_{t,i})_c, \quad \hat{y}_t = \arg \max_c \bar{p}_{t,c}. \quad (4)$$

This yields exactly one prediction per object, from which TIC-level accuracy and macro F_1 are computed. Figure 4 illustrates why the aggregation matters: an individual segment of an eclipsing binary may lack a clear secondary eclipse and be misclassified, while the probability average over all segments remains correct.

We report both segment-level and TIC-level metrics, but use TIC-level macro F_1 on the validation set as the sole criterion for checkpoint selection, since it corresponds directly to the way the model would be applied to unlabelled data.

4.5. Backbones

Three backbones are supported within the same architecture:

Transformer Standard encoder blocks trained from scratch, implemented in PyTorch (Paszke et al. 2019).

Chronos-2 A 120M-parameter encoder-only time series foundation model developed by Amazon (Ansari et al. 2025). It was pretrained for probabilistic forecasting on generic time series and performs its own tokenisation and patching, so it consumes the raw normalised flux.

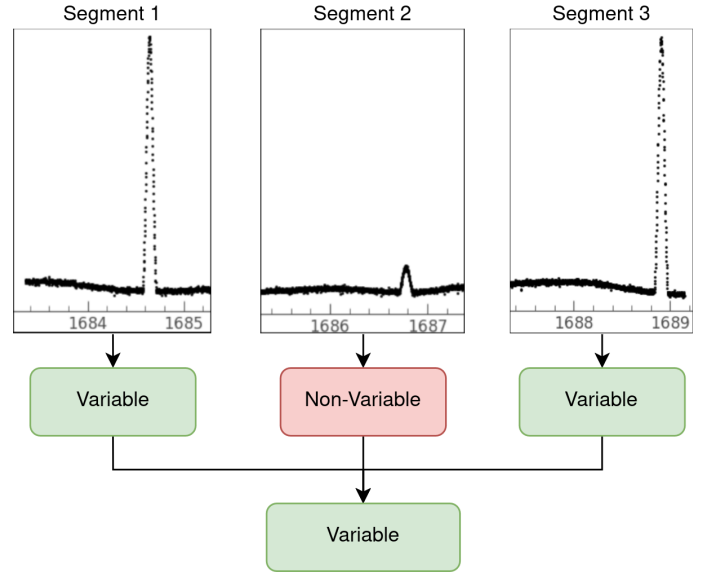


Fig. 4. Segment-level versus object-level classification. One of the three segments of this eclipsing binary is misclassified as non-variable because its secondary eclipse is poorly defined; averaging class probabilities over segments recovers the correct label.

Qwen 2.5 0.5B A 500M-parameter pretrained large language model (Yang et al. 2025), adapted to accept patch embeddings in place of text tokens.

We did not evaluate pretrained vision transformers, since the problem is one-dimensional at the physical level and rendering light curves as images discards information for no obvious benefit.

Both pretrained backbones were adapted with LoRA (Hu et al. 2021), with the original weights frozen and rank-8 adapters ($\alpha = 16$) trained alongside the embedding, fusion, and classification heads.

4.6. Training protocol

A standard train/validation/test split at the TIC level was used, with the test set touched only once, for the final model. We also implemented k -fold cross-validation, but it multiplied training time by $\sim k$ without a measurable improvement in validation performance, so it was not adopted.

Hyperparameters were fixed after an initial exploration phase and held constant across all comparisons (Table 3). Patch size proved the most influential single parameter: values from 4 to 64 were tested, and 16 was clearly optimal, with much larger or smaller values either overfitting or failing to converge. The transformer width parameters showed a threshold behaviour – below a minimum viable value the model does not converge, above it further capacity brings no gain. Cosine annealing of the learning rate outperformed both a flat schedule and the other schedulers tested. Balanced sampling, tried on the imbalanced Task B, caused over-prediction of the eclipsing class and was abandoned in favour of the weighting in Eq. (3).

5. Results

5.1. Experiment overview

Sixteen configurations were trained on both tasks under the protocol of Sect. 4. The baseline replaces the transformer blocks

Table 3. Training configuration. Values in parentheses apply to the fine-tuned models.

Parameter	Value
Epochs	50 (10 for fine-tunes)
Learning rate	5×10^{-5} , cosine annealed to 10^{-6}
Optimiser	AdamW
Weight decay	1×10^{-2}
Batch size	64 (32 for Qwen 2.5)
Patch size	16 (Chronos-2 consumes raw flux)
Dropout	0.15
n_{head}	8
n_{layers}	4
d_{model}	256
d_{ff}	1024
d_{fusion}	256
LoRA r / α	8 / 16
Loss	Eq. (3)

Table 4. Validation TIC-level macro F_1 for all evaluated configurations. The best result per task is in bold.

Model	Backbone	Task A	Task B
Baseline	CNN	0.9061	0.9232
2-min view	Transformer	0.8396	0.6618
8-min view	Transformer	0.8704	0.8223
32-min view	Transformer	0.8684	0.8614
128-min view	Transformer	0.8436	0.9110
4-channel	Transformer	0.8931	0.9230
6-channel	Transformer	0.8983	0.9240
Chronos-2	TSFM fine-tune	0.9261	0.9496
Qwen 2.5 0.5B	LLM fine-tune	0.9129	0.9139

with two 1D convolutional layers (kernel size 3) while keeping the multi-branch architecture, preprocessing, loss, and aggregation identical, so that the comparison isolates the sequence encoder. Table 4 lists validation TIC-level macro F_1 for all configurations.

Three results stand out. First, single-view transformers are consistently worse than the baseline, and the ranking of views is task-dependent in exactly the way the physics predicts: the 2-min view is competitive on Task A but collapses on Task B ($F_1 = 0.66$), because separating rotating from eclipsing variables requires timescales of days, which a 8.5-hour window cannot resolve. Conversely the 128-min view is the best single view on Task B and among the worst on Task A.

Second, combining views recovers most of the gap: the 4- and 6-channel transformers reach the baseline on Task B and come close on Task A. Since the baseline CNN has access to the same six channels, the small residual difference indicates that the multi-resolution representation and the segmentation scheme – not the choice of sequence encoder – account for most of the achievable performance.

Third, the fine-tuned Chronos-2 model is the only configuration that exceeds the baseline on both tasks, by $\Delta F_1 = 0.020$ and 0.026 respectively. The fine-tuned LLM outperforms every from-scratch transformer on Task A but does not beat the baseline on either task, despite being four times larger than Chronos-2 – consistent with the expectation that pretraining on time series transfers better to photometric data than pretraining on text.

We attempted to improve on Chronos-2 by further tuning: increasing dropout, varying the LoRA rank and scaling, test-

Table 5. Test-set performance of the fine-tuned Chronos-2 model.

Task A – accuracy 0.870, macro F_1 0.869, TIC accuracy 0.895, TIC macro F_1 0.893				
Class	Precision	Recall	F_1	Support
N (non-variable)	0.85	0.97	0.90	259
V (variable)	0.96	0.82	0.88	243
Task B – accuracy 0.900, macro F_1 0.834, TIC accuracy 0.933, TIC macro F_1 0.880				
Class	Precision	Recall	F_1	Support
E (eclipsing)	0.83	0.67	0.74	15
P (pulsating)	0.94	0.99	0.96	81
R (rotating)	0.94	0.93	0.94	39

ing additional channel combinations, and enlarging the fusion head. None of these produced a configuration better than the original, which was therefore adopted as final. Training took approximately 100 min (Task A) and 30 min (Task B). The checkpoints selected by validation TIC-level macro F_1 were epoch 5 for Task A and epoch 4 for Task B (Fig. 5); in both cases the validation loss had not yet begun to rise and the training loss was only marginally lower, indicating that the selected checkpoints are not significantly overfitted.

5.2. Test-set performance

Table 5 reports the performance of the final model on the held-out test set, which was used exactly once.

On Task A the TIC-level macro F_1 drops by ≈ 0.02 relative to validation, which is a reasonable generalisation gap for a dataset of this size. The composition of the score changes, however: for the variable class, precision rises and recall falls substantially compared with validation. The classifier has become conservative.

On Task B the drop is larger, $\Delta F_1 \approx 0.07$. It is almost entirely attributable to the eclipsing class, whose recall of 0.67 corresponds to five misclassified objects out of fifteen. The pulsating and rotating classes generalise essentially without loss, with test metrics within 0.01 of their validation values.

6. Discussion

6.1. Failure modes

We inspected each of the five misclassified eclipsing systems individually (Table 6). Every one is an atypical member of its class. Two are ELL systems whose ellipsoidal maxima are smooth rather than sharp, making them morphologically indistinguishable from spotted rotating variables at the available signal-to-noise; one EW system has a continuously varying light curve resembling a pulsator; one EA system is too noisy for its eclipses to be recovered; and one EA system has an atypically long period, eclipsing roughly once per 50 d, so that most 23-day segments contain no eclipse at all (Fig. 6).

The last case exposes a structural limitation of segment-based aggregation by mean probability: for objects whose defining feature occurs in a small minority of segments, averaging dilutes the signal. A max-probability or attention-weighted aggregation would address this, at the cost of increased sensitivity to single-segment false positives. The two ELL cases are a different matter – the distinction between ellipsoidal variation

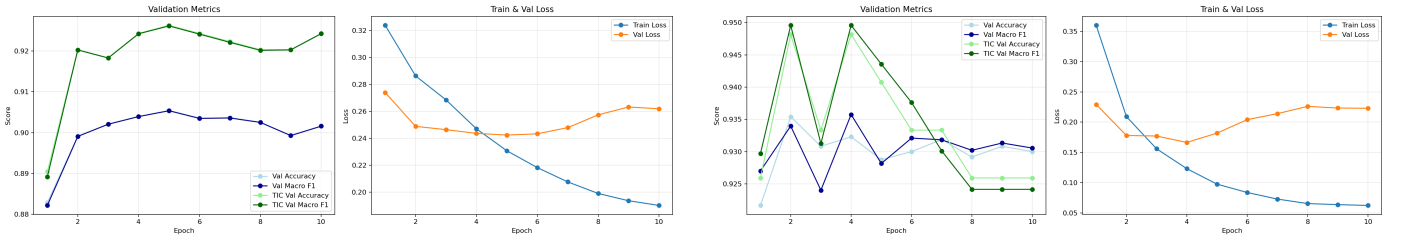


Fig. 5. Training and validation curves of the fine-tuned Chronos-2 model for Task A (left) and Task B (right). The selected checkpoints are epoch 5 and epoch 4 respectively, just before the validation loss begins to diverge from the training loss.

Table 6. The five misclassified eclipsing systems in the Task B test set.

TIC	Subtype	Remark
224595426	EA	Very noisy; eclipses not clearly identifiable.
229710150	ELL	Smooth maxima; resembles a rotating variable.
259238257	ELL	Smooth maxima; resembles a rotating variable.
288346083	EA	Atypically long period, one eclipse per ~ 50 d.
362226108	EW	Continuous variation resembling a pulsator.

plete pipeline can process a million light curves per day on a single consumer workstation. Neither figure reflects a tuned implementation; datacentre GPUs and wider parallelism would improve both substantially.

6.4. Applicability

This work forms part of an ongoing effort at the Astronomical Institute of the Czech Academy of Sciences to build automated classifiers for the identification of interesting objects, exoplanets in particular. Existing transit-search pipelines produce large numbers of false positives, typically variable stars misidentified as planet hosts. Because transits resemble the eclipses of a detached binary far more than they resemble pulsation or rotational modulation, a Task B classifier that reliably recognises pulsating and rotating variables can remove a substantial fraction of these false positives before human vetting.

The timing is favourable: Gaia DR4 (Gaia Collaboration et al. 2016) and PLATO (Rauer et al. 2025) will both deliver large volumes of photometric time series in the near future. Models of the kind presented here can either contribute directly to their filtering or serve as a starting point for survey-specific classifiers.

6.5. Limitations and outlook

The principal limitation is the size of the labelled sample, and in particular the 101 eclipsing systems, which is what ultimately bounds Task B performance. Three directions follow naturally. First, unsupervised pretraining on the millions of available unlabelled TESS light curves would let the model learn the structure of photometric time series without labels – the domain gap that Chronos-2 must currently bridge from generic time series. Second, modern architectural components – gated feed-forward networks, FFT-based attention, and learned rather than linear fusion, including gating at the branch level – are straightforward additions to the present design. Third, and most directly, a larger labelled catalogue, particularly of eclipsing systems, would help more than any architectural change.

7. Conclusions

We have evaluated transformer-based classifiers of stellar variability on 3347 manually labelled TESS light curves, comparing transformers trained from scratch, fine-tuned pretrained models, and a convolutional baseline under an identical protocol. Our main findings are:

1. Representing each light-curve segment by multiple time-domain views at different cadences is the decisive design

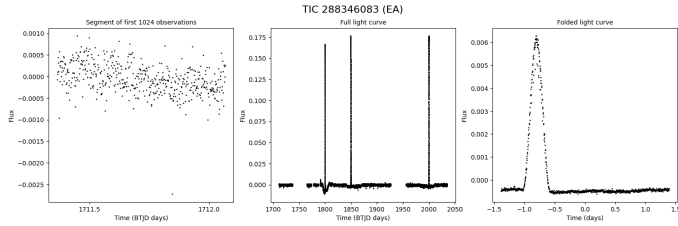


Fig. 6. TIC 288346083, a misclassified EA system with an atypically long period. Most segments contain no eclipse, so the object-level probability average is dominated by featureless windows.

and rotational modulation is genuinely marginal from photometry alone and would challenge a human classifier as well.

6.2. Operating point

The precision–recall asymmetry on Task A has practical consequences. A classifier with 0.96 precision and 0.82 recall on the variable class is well suited to building large, high-purity samples of variable stars from unlabelled archives – for instance as a corpus for unsupervised pretraining – but will miss roughly one variable in five. For applications where completeness matters more than purity, the decision threshold on $\bar{p}_{t,c}$ should be adjusted accordingly; the aggregation of Eq. (4) provides a calibrated probability at which to do so.

6.3. Throughput

Survey-scale application requires the pipeline to process $\sim 10^6$ light curves in reasonable time. There are two potential bottlenecks. Preprocessing runs at ≈ 13.8 light curves s^{-1} on an 8-core AMD Ryzen 5 9700X, i.e. about 1.2×10^6 per day; inference on the fine-tuned Chronos-2 model runs at ≈ 12.6 light curves s^{-1} on a consumer AMD Radeon RX 9070 GPU, about 1.1×10^6 per day. The two stages are therefore well balanced, and the com-

choice. Single-view transformers underperform the baseline on both tasks, and the failure is systematic: high-cadence views cannot resolve day-scale morphology and low-cadence views cannot resolve pulsation. Combining four views closes most of the gap.

2. Fine-tuning a time series foundation model transfers effectively to astronomical photometry. Chronos-2 with LoRA adapters was the only configuration to beat the convolutional baseline on both tasks, reaching validation TIC-level macro F_1 of 0.926 and 0.950, and test values of 0.893 and 0.880.
3. Pretraining domain matters more than model size. The 500M-parameter LLM did not match the 120M-parameter TSFM, despite being four times larger.
4. Statistical patch embeddings, which augment each patch with its mean, dispersion, extrema, slope, and valid fraction, were worth 0.05–0.10 in macro F_1 – the largest single architectural gain we measured.
5. The residual errors are dominated by genuinely atypical objects. All five misclassified eclipsing systems in the test set are non-standard, and two of them are arguably ambiguous from photometry alone.

The complete pipeline sustains $\sim 10^6$ light curves per day on a single consumer workstation, which makes it directly applicable to survey-scale archives and to the reduction of false positives in transit searches, ahead of the Gaia DR4 and PLATO data releases.

Acknowledgements. We thank M. Skarka for providing the cleaned labelled dataset. This paper includes data collected by the TESS mission, obtained from the Mikulski Archive for Space Telescopes (MAST). Funding for the TESS mission is provided by the NASA Explorer Program.

Data availability

The training code, configuration files, and experiment logs are available at <https://github.com/>. The labelled catalogue is published by Skarka et al. (2022); the underlying TESS light curves are publicly available from MAST.

References

- Ansari, A. F., Shchur, O., Küken, J., et al. 2025, Chronos-2: From Univariate to Universal Forecasting
- Bellm, E. C., Kulkarni, S. R., Barlow, T., et al. 2019, PASP, 131, 068003
- Borucki, W. J., Koch, D., Basri, G., et al. 2010, Science, 327, 977
- Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1
- Hu, E. J., Shen, Y., Wallis, P., et al. 2021, LoRA: Low-Rank Adaptation of Large Language Models
- Hubble, E. 1929, Proceedings of the National Academy of Sciences, 15, 168
- Lomb, N. R. 1976, Ap&SS, 39, 447
- MAST. 2024, TESS Data Products Description, Mikulski Archive for Space Telescopes, <https://outerspace.stsci.edu/display/TESS/>
- Paszke, A., Gross, S., Massa, F., et al. 2019, in Advances in Neural Information Processing Systems, Vol. 32 (Curran Associates, Inc.)
- Rauer, H., Aerts, C., Cabrera, J., et al. 2025, Experimental Astronomy, 59, 26
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2014, Journal of Astronomical Telescopes, Instruments, and Systems, 1, 014003
- Scargle, J. D. 1982, ApJ, 263, 835
- Skarka, M., Žák, J., Fedurco, M., et al. 2022, A&A, 666, A142
- Udalski, A., Szymański, M. K., Soszyński, I., & Poleski, R. 2008, The Optical Gravitational Lensing Experiment. Final Reductions of the OGLE-III Data
- Vaswani, A., Shazeer, N., Parmar, N., et al. 2017, in Advances in Neural Information Processing Systems, Vol. 30 (Curran Associates, Inc.)
- Yang, A., Yang, B., Zhang, B., et al. 2025, Qwen2.5 Technical Report